SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	Kishore Gadhi	**Reg. No.:**	192365051
Section / Batch:		**Date:**	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – APPLICATION BASED QUESTIONS

Q1. A social media system analyzes posts using the recurrence T(n) = 2T(n/2) + n log n. Apply the Master Theorem to determine the time complexity. Identify parameters and analyze the space complexity.

Q2. A smart grid system processes energy distribution data using the recurrence T(n) = 4T(n/2) + n log n. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the case selection. Analyze how the system scales with increasing network size. Interpret the result using asymptotic notation.

Code of Conduct

I ____________________________(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Q01). **Social Media Trend Analysis System**

A social media system analyzes posts using the recurrence T(n) = 2T(n/2) + n log n. Apply the Master Theorem to determine the time complexity. Identify parameters and analyze the space complexity.

Ans):

Social Media Trend Analysis System

Given the recurrence:

$$T(n) = 2T(n/2) + n\log n$$

We solve it using the **Master Theorem**.

Step 1: Compare with Master Theorem

The standard form of the Master Theorem is

$$T(n) = aT(n/b) + f(n)$$

Comparing,

- a=2
- b=2
- f(n)= n log n

Step 2: Compute the Critical Function

Calculate

n^logb(a)

Since

a=2, b=2

log2(2) = 1

Therefore,

n^logb(a) = n^1 = n

Step 3: Compare (f(n)) with (n^{\log_b a})

Given

$f(n)=n\log n$

Since

$n\log n=\Theta(n\log^1 n)$

This matches

$f(n)=\Theta(n^{\log_b a} \log^k n)$

where

- $(k=1)$

Step 4: Apply Master Theorem

Master Theorem **Case 2** states:

If

$f(n)=\Theta(n^{\log_b a} \log^{k+1} n)$

then

$T(n)= \Theta (n^{\log_b a} \log^{k+1}(n))$

Substituting

- $n^{\log_b a}=n$
- $k=1$

we get

$T(n)= \Theta\, n \log^2 n$

Step 5: Final Time Complexity

Big- Θ (Tight Bound)

$\Theta (n \log^2 n)$

Big-O (Upper Bound)

Since

$T(n)= \Theta (n \log^2 n)$

$O(n \log^2 n)$

Big-Omega (Lower Bound)

Similarly,

$$\Omega(n \log^2 n)$$

Hence,

$$\{T(n)= \Theta (n \log^2 n)$$

Step 6: Space Complexity Analysis

Each recursive call divides the problem size by 2.

Recursion depth:

$$n/2^k=1$$

Taking logarithm,

$$k=\log_2 n$$

Thus, there are

$$\log n$$

recursive levels.

Each level uses only **constant extra memory** (excluding the recursive calls).

Therefore,

Space Complexity = Θ (log n).

Q.3) **Smart Grid Energy Distribution System**

A smart grid system processes energy distribution data using the recurrence T(n) = 4T(n/2) + n log n. Apply the Master Theorem to determine the time complexity. Clearly identify parameters and justify the case selection. Analyze how the system scales with increasing network size. Interpret the result using asymptotic notation.

ANS:

Smart Grid Energy Distribution System

Given the recurrence:

$$T(n) = 4T(n/2) + n\log n$$

We solve it using the Master Theorem.

Step 1: Compare with the Master Theorem

The standard form of the Master Theorem is

$$T(n) = aT(n/b) + f(n)$$

Comparing,

- a = 4
- b = 2
- f(n) = n log n

Step 2: Compute the Critical Function

Calculate

n ^ log_b (a)

Since

a=4, b=2

log_2 (4) = 2

Therefore,

n ^ log_b a = n^2

Step 3: Compare f(n) with n^log_b(a)

Given

f(n) = n log n

We compare:

- $n \log n$
- n^2

Since

$$n \log n = O(n^{(2-\varepsilon)})$$

Choose

$$\varepsilon = 1/2$$

because

$$n \log n = O(n^{3/2})$$

Thus,

$$f(n) = O(n^{\log_b a - \varepsilon})$$

Step 4: Justify the Master Theorem Case

The Master Theorem Case 1 states:

If

$$f(n) = O(n^{\log_b a - \varepsilon})$$

for some constant $\varepsilon > 0$, then

$$T(n) = \Theta(n^{\log_b a})$$

Since

- $n^{\log_b a} = n^2$
- $f(n) = n \log n$ grows asymptotically slower than n^2,

Case 1 applies.

Step 5: Determine the Time Complexity

Applying Case 1,

$$T(n) = \Theta(n^2)$$

Step 6: Asymptotic Notation

Big-Theta (Tight Bound)

$$\Theta(n^2)$$

Big-O (Upper Bound)

$$O(n^2)$$

Big-Omega (Lower Bound)

$$\Omega(n^2)$$

Hence,

$$T(n) = \Theta(n^2)$$

Step 7: Space Complexity Analysis

The recurrence divides the problem size by 2 at each recursive level.

Recursion depth:

$$n/2^k = 1$$

Taking logarithm,

$$k = \log_2 n$$

Thus, the recursion stack contains

$$\Theta(\log n)$$

levels.

Assuming each recursive call requires constant extra memory,

$$\text{Space Complexity} = \Theta(\log n)$$

Step 8: Scalability Analysis

- The system recursively processes 4 subproblems, each of size $(n/2)$.
- The cost of dividing and combining results is only $n \log n$, which is much smaller than the recursive computation.
- Therefore, the recursive work dominates the overall execution time.

As the network size increases:

- Doubling the number of nodes approximately quadruples the execution time because the algorithm has quadratic growth.

- **Hence, the system scales reasonably for moderate-sized smart grids but becomes computationally expensive for very large networks.**

Using the Master Theorem:

- $A = 4$, $b = 2$, $f(n) = n \log n$
- **$n^{\log_b a} = n^2$**
- **Since $f(n) = n \log n = O(n^{2-\varepsilon})$ for some $\varepsilon > 0$, Master Theorem Case 1 applies.**
- **Therefore,**

$$T(n) = \Theta(n^2)$$